Guidance and Control of Unmanned Autonomous Vehicles

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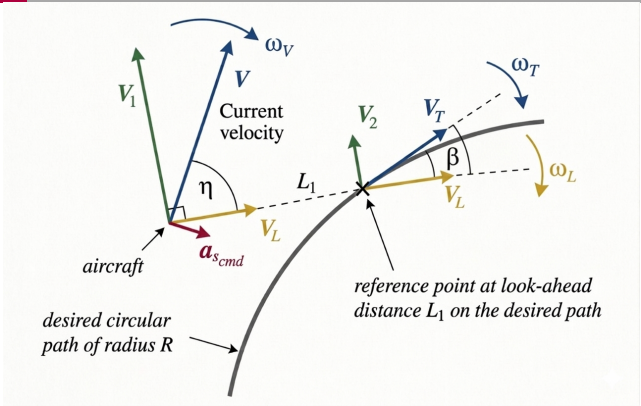
Circular Path using Nonlinear Guidance Method

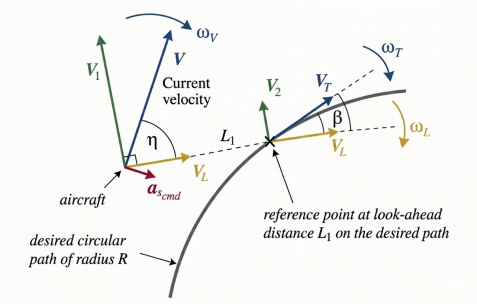


Assumptions

- All angles and angular velocities are positive in clockwise direction
- Inner-loop control system is sufficiently fast.
- Distance L_1 is constant.

- UAV velocity (V) components: $\boxed{V_L = V\cos\eta, \quad V_1 = V\sin\eta}$
- Direction of V_T: Tangent to the path
- **Reference point velocity** (Components of V_T): $\boxed{V_L = V_T\cos\beta, \quad V_2 = V_T\sin\beta}$

Circular Path: Analysis of the Nonlinear Guidance Method



- Radius of circular path: R

$$V_T = \frac{V_L}{\cos\beta} = \frac{V\cos\eta}{\cos\beta}$$

$$V_2 = V_T \sin\beta = V\cos\eta\tan\beta$$

- Rate of change of angles η and β

$$\dot{\eta} = \omega_L - \omega_V, \quad \dot{\beta} = \omega_L - \omega_T$$

- Angular velocities ω_L and ω_T

$$\omega_L = \frac{V_1 - V_2}{L_1} = \frac{V}{L_1}(\sin\eta - \cos\eta\tan\beta)$$

$$\omega_T = \frac{V_T}{R} = \frac{V\cos\eta}{R\cos\beta}$$

- As $a_s = a_{s_{\text{cmd}}} = \dfrac{2V^2}{L_1}\sin\eta$

$$\omega_V = \frac{a_s}{V} = \frac{2V}{L_1}\sin\eta$$

Circular Path: Analysis of the Nonlinear Guidance Method

- Dynamics η and β

$$\dot{\eta} = -\frac{V}{L_1}(\sin\eta + \cos\eta\tan\beta)$$
$$\dot{\beta} = \frac{V}{L_1}(\sin\eta - \cos\eta\tan\beta) - \frac{V\cos\eta}{R\cos\beta}$$

- **Assumptions**: $|\eta(t)| < 90°, \quad |\beta(t)| < 90°$
- Differential equations are well-behaved and satisfy Lipschitz Condition.
- $\eta(0)$: Initial approach angle of UAV to ensure reference point ahead of vehicle
- $\beta(0)$: Movement of reference point along desired path in forward direction

- How to obtain stationary points η_0 and β_0?
- Equate $\dot{\eta} = 0$ and $\dot{\beta} = 0$

$$\boxed{\eta_0 = -\beta_0}$$

$$\frac{V}{L_1}(\sin\eta_0 + \sin\eta_0) - \frac{V}{R} = 0$$
$$\implies \frac{2V}{L_1}\sin\eta_0 - \frac{V}{R} = 0$$

$$\boxed{\eta_0 = \sin^{-1}\left(\frac{L_1}{2R}\right) = -\beta_0}$$

- Look ahead distance $L_1 \leq 2R$

Linearization About a Circular Trajectory

- Small perturbation from stationary point: $\Delta\eta$ and $\Delta\beta$

$$\eta = \eta_0 + \Delta\eta, \quad \beta = \beta_0 + \Delta\beta$$

- Dynamics of $\Delta\eta$ using the Taylor's series

$$\Delta\dot{\eta} \approx -\frac{V}{L_1}\left[(\cos\eta_0 - \sin\eta_0 \tan\beta_0)\Delta\eta + (\cos\eta_0 \sec^2\beta_0)\Delta\beta\right]$$

- At stationary points,

$$\sin\eta_0 = -\sin\beta_0 = \frac{L_1}{2R}, \qquad \cos\eta_0 = \cos\beta_0 = \sqrt{1 - \left(\frac{L_1}{2R}\right)^2}$$

- On simplifications of the dynamics of variable $\Delta\eta$

$$\Delta\dot{\eta} \approx -\frac{V}{L_1\sqrt{1 - \left(\frac{L_1}{2R}\right)^2}}(\Delta\eta + \Delta\beta)$$

Linearization About a Circular Trajectory

- Dynamics of $\Delta\dot{\beta}$ using the Taylor's series

$$\Delta\dot{\beta} \approx \frac{V}{L_1}\left[\cos\eta_0 + \sin\eta_0\tan\beta_0 + \frac{L_1}{R}\sin\eta_0\sec\beta_0\right]\Delta\eta - \frac{V}{R}\left[\frac{R}{L_1}\cos\eta_0\sec^2\beta_0 + \cos\eta_0\sec\beta_0\tan\beta_0\right]\Delta\beta$$

- On simplifying the above equation

$$\Delta\dot{\beta} \approx \frac{V}{L_1\sqrt{1-\left(\frac{L_1}{2R}\right)^2}}\left[\Delta\eta + \left\{2\left(\frac{L_1}{2R}\right)^2 - 1\right\}\Delta\beta\right]$$

- Undamped natural frequency ω_n and damping ratio ζ

$$\omega_n = \frac{\sqrt{2}V}{L_1}$$

$$\zeta = \frac{1}{\sqrt{2}}\sqrt{1-\left(\frac{L_1}{2R}\right)^2}$$

- Any observation as compared to the straight-line path?
- ω_n remains unchanged but ζ changed as a function of L_1 and R.

Linearization About a Circular Trajectory

- Linearized dynamics in compact form

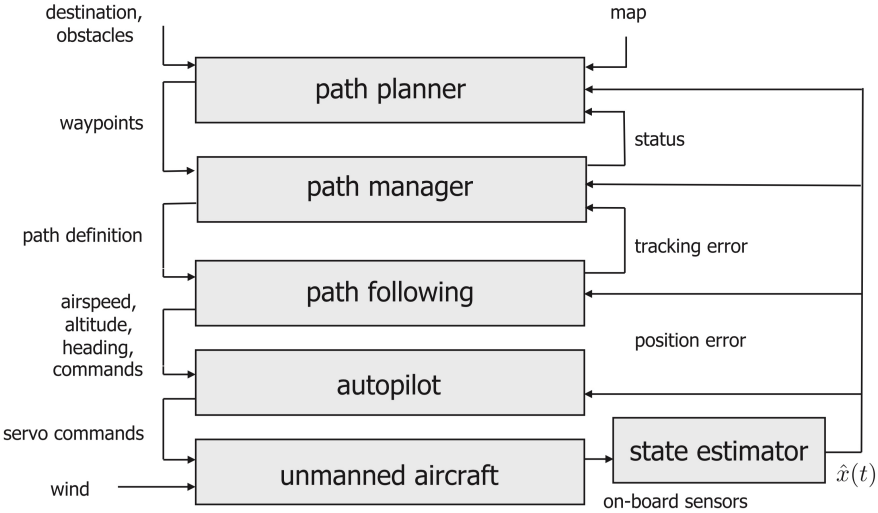
$$\begin{bmatrix} \Delta\dot{\eta} \\ \Delta\dot{\beta} \end{bmatrix} = \begin{bmatrix} -\dfrac{V}{L_1\sqrt{1-\left(\frac{L_1}{2R}\right)^2}} & -\dfrac{V}{L_1\sqrt{1-\left(\frac{L_1}{2R}\right)^2}} \\ \dfrac{V}{L_1\sqrt{1-\left(\frac{L_1}{2R}\right)^2}} & \dfrac{V\left[2\left(\frac{L_1}{2R}\right)^2-1\right]}{L_1\sqrt{1-\left(\frac{L_1}{2R}\right)^2}} \end{bmatrix} \begin{bmatrix} \Delta\eta \\ \Delta\beta \end{bmatrix}$$

- Characteristic equation

$$s^2 + \left(\frac{V}{L_1\sqrt{1-\left(\frac{L_1}{2R}\right)^2}} - \frac{V\left[2\left(\frac{L_1}{2R}\right)^2-1\right]}{L_1\sqrt{1-\left(\frac{L_1}{2R}\right)^2}}\right)s - \frac{V^2\left[2\left(\frac{L_1}{2R}\right)^2-1\right]}{L_1^2\left(1-\left(\frac{L_1}{2R}\right)^2\right)} + \frac{V^2}{L_1^2\left(1-\left(\frac{L_1}{2R}\right)^2\right)} = 0$$

$$s^2 + \frac{2V}{L_1\sqrt{1-\left(\frac{L_1}{2R}\right)^2}}\left[1-\left(\frac{L_1}{2R}\right)^2\right]s + \frac{2V^2}{L_1^2} = 0 \implies \boxed{s^2 + \frac{2V}{L_1}\sqrt{1-\left(\frac{L_1}{2R}\right)^2}\,s + \frac{2V^2}{L_1^2} = 0}$$

Control Architecture



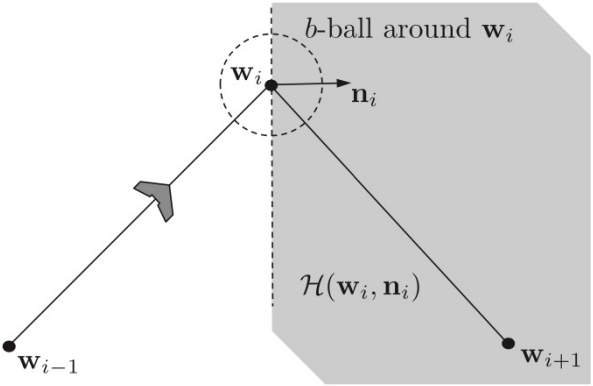
UAV Path Manager

- Why are the straight path and circular orbit paths important?
- How to generate a general path using straight line paths and circular orbits?

- Combination of these paths to synthesize a general class of paths to follow a series of waypoints.
- **Waypoint path**: An ordered sequence of waypoints

$$\mathcal{W} = \{\boldsymbol{w}_1, \boldsymbol{w}_2, \cdots, \boldsymbol{w}_N\}$$

$$\boldsymbol{w}_i = (w_{n,i}, w_{e,i}, w_{d,i})^\top \in \mathbb{R}^3$$



UAV Path Manager

- UAV tracking $\overline{\boldsymbol{w}_{i-1}\boldsymbol{w}_i}$ wants to switch to the path $\overline{\boldsymbol{w}_i\boldsymbol{w}_{i+1}}$
- What would be the best method for determining whether the UAV has reached a waypoint w_i?
- Switch of strategy when the UAV enters a ball of the desired radius b

$$\|\boldsymbol{p}(t) - \boldsymbol{w}_i\| \leq b,$$

where b is the size of ball and $\boldsymbol{p}(t)$ is the location of UAV.

- **Issues with this approach**:
  - Presence of disturbances, like wind
  - Very small b
  - Short segment from $\boldsymbol{w}_{i-1}$ to $\boldsymbol{w}_i$
  - No time to converge by tracking algorithm (UAV never enters the b-ball around $\boldsymbol{w}_i$)

UAV Path Manager

- How to avoid such issues to obtain a smooth path to be tracked?
- Half-plane switching criteria avoid such sensitivity
- What do you mean by the half-plane?

Definition of Half Plane

Given a vector $\boldsymbol{r} \in \mathbb{R}^3$ and $\boldsymbol{n} \in \mathbb{R}^3$, a half-plane can be defined as

$$\boxed{\mathcal{H}(\boldsymbol{r}, \boldsymbol{n}) \triangleq \{\boldsymbol{p} \in \mathbb{R}^3 : (\boldsymbol{p} - \boldsymbol{r})^\top \boldsymbol{n} \geq 0\}}$$

- Unit vector along the line $\overline{\boldsymbol{w}_i \boldsymbol{w}_{i+1}}$:

$$\boldsymbol{q}_i \triangleq \frac{\boldsymbol{w}_{i+1} - \boldsymbol{w}_i}{\|\boldsymbol{w}_{i+1} - \boldsymbol{w}_i\|}$$

- Unit normal to 3-D half plane that separates the line $\overline{\boldsymbol{w}_{i-1} \boldsymbol{w}_i}$ from the line $\overline{\boldsymbol{w}_i \boldsymbol{w}_{i+1}}$

$$\boldsymbol{n}_i \triangleq \frac{\boldsymbol{q}_{i-1} + \boldsymbol{q}_i}{\|\boldsymbol{q}_{i-1} + \boldsymbol{q}_i\|}$$

- Switching of the vehicle's guidance strategy when it enters $\mathcal{H}(\boldsymbol{w}_i, \boldsymbol{n}_i)$

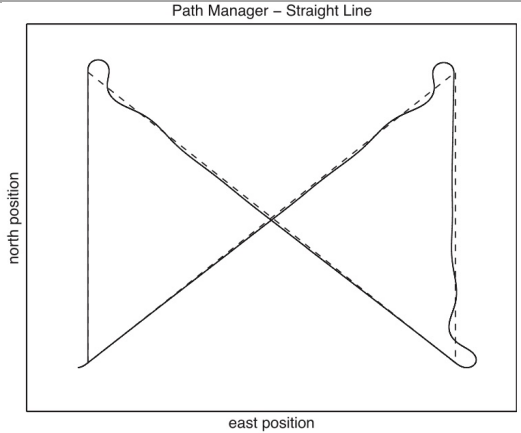
UAV Path Manager

Algorithm 5 Follow Waypoints: $(\mathbf{r}, \mathbf{q}) = \text{followWpp}(\mathcal{W}, \mathbf{p})$

Input: Waypoint path $\mathcal{W} = \{\mathbf{w}_1, \ldots, \mathbf{w}_N\}$, MAV position $\mathbf{p} = (p_n, p_e, p_d)^\top$.

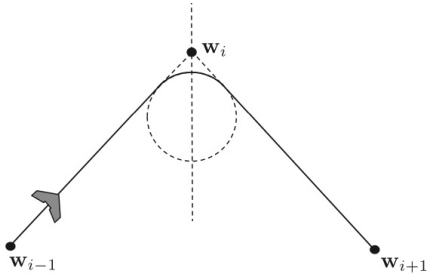
Require: $N \geq 3$

```
if New waypoint path W is received then
    Initialize waypoint index: i ← 2
end if
r ← w_{i-1}
q_{i-1} ← (w_i − w_{i-1}) / ||w_i − w_{i-1}||
q_i ← (w_{i+1} − w_i) / ||w_{i+1} − w_i||
n_i ← (q_{i-1} + q_i) / ||q_{i-1} + q_i||
if p ∈ H(w_i, n_i) then
    Increment i ← (i + 1) until i = N − 1
end if
return r, q = q_{i-1} at each time step
```



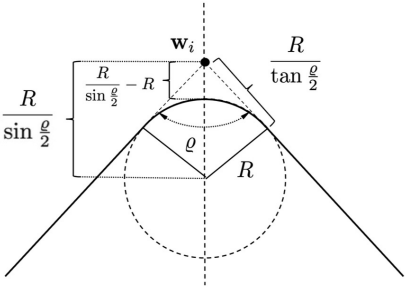
- Simple algorithm to reach waypoint, but **non-smooth, and not balanced transition**
- **What could be other possible ways to generate smoother paths?**

UAV Path Manager: Smooth transition between different path segments



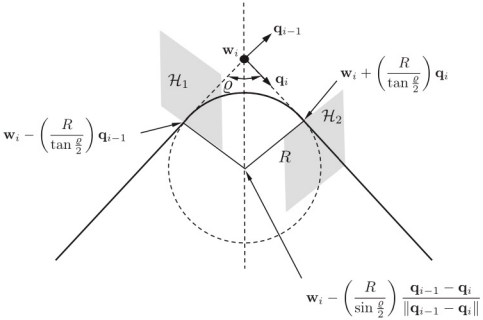
- Are there any issues with such transitions?
- **Waypoint not visited by the UAV**
  - **Angle between $\overline{\boldsymbol{w}_{i-1}\boldsymbol{w}_i}$ and $\overline{\boldsymbol{w}_i\boldsymbol{w}_{i+1}}$?**

- Distances between center and location where the fillet intersects $\overline{\boldsymbol{w}_i}$ and $\boldsymbol{w}_{i+1}$ from waypoint



$$\boxed{\varrho \triangleq \cos^{-1}\left(-\boldsymbol{q}_{i-1}^{\top}\boldsymbol{q}_i\right)}$$

UAV Path Manager: Fillet Maneuvers using Path Following Algorithm



- Follow straight-line $\overline{\boldsymbol{w}_{i-1}\boldsymbol{w}_i}$ until entering $\mathcal{H}_1$
- Right-handed orbit of radius R until entering $\mathcal{H}_2$
- Follow straight-line segment $\overline{\boldsymbol{w}_i\boldsymbol{w}_{i+1}}$

- Center of the fillet

$$\boxed{\boldsymbol{c} = \boldsymbol{w}_i - \left(\frac{R}{\sin(\varrho/2)}\right)\frac{\boldsymbol{q}_{i-1}-\boldsymbol{q}_i}{\|\boldsymbol{q}_{i-1}-\boldsymbol{q}_i\|}}$$

- Location of the half-plane $\mathcal{H}_1$ with normal vector $\boldsymbol{q}_{i-1}$

$$\boxed{\boldsymbol{r}_1 = \boldsymbol{w}_i - \left(\frac{R}{\tan(\varrho/2)}\right)\boldsymbol{q}_{i-1}}$$

- Location of the half-plane $\mathcal{H}_2$ with normal vector $\boldsymbol{q}_i$

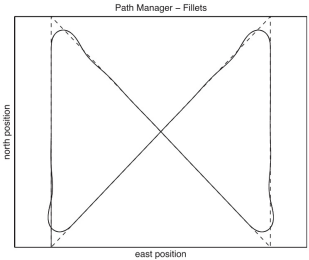
$$\boxed{\boldsymbol{r}_2 = \boldsymbol{w}_i + \left(\frac{R}{\tan(\varrho/2)}\right)\boldsymbol{q}_i}$$

UAV Path Manager

Algorithm 6 Follow Waypoints with Fillets: (flag, $\mathbf{r}$, $\mathbf{q}$, $\mathbf{c}$, ρ, λ) = followWppFillet($\mathcal{W}$, $\mathbf{p}$, R)

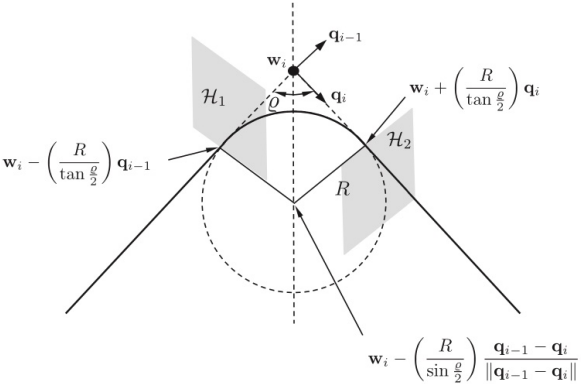
Input: Waypoint path $\mathcal{W} = \{\mathbf{w}_1, \ldots, \mathbf{w}_N\}$, MAV position $\mathbf{p} = (p_n, p_e, p_d)^\top$, fillet radius R.
Require: $N \geq 3$

```
if New waypoint path W is received then
    Initialize waypoint index: i ← 2, and state machine: state ← 1.
end if
q_{i-1} ← (w_i − w_{i-1}) / ||w_i − w_{i-1}||
q_i ← (w_{i+1} − w_i) / ||w_{i+1} − w_i||
ρ ← cos^{-1}(−q_{i-1}^T q_i)
if state = 1 then
    flag ← 1
    r ← w_{i-1}
    q ← q_{i-1}
    z ← w_i − (R / tan(ρ/2)) q_{i-1}
    if p ∈ H(z, q_{i-1}) then
        state ← 2
    end if
else if state = 2 then
    flag ← 2
    c ← w_i − (R / sin(ρ/2)) (q_{i-1} − q_i) / ||q_{i-1} − q_i||
    ρ ← R
    λ ← sign(q_{i-1,n} q_{i,e} − q_{i-1,e} q_{i,n})
    z ← w_i + (R / tan(ρ/2)) q_i
    if p ∈ H(z, q_i) then
        i ← (i + 1) until i = N − 1.
        state ← 1
    end if
end if
return flag, r, q, c, ρ, λ.
```



- Disadvantage: Change in path length with the insertion of fillets
- **Cooperative timing problems**: High-quality estimate of the path length, or time required to traverse a certain waypoint path

UAV Path Manager: Total Path Length Change



- Total path length between waypoints without fillets

$$|W| \triangleq \sum_{i=2}^{N} \|\boldsymbol{w}_i - \boldsymbol{w}_{i-1}\|$$

- What would be total path length $|W|_F$ with the fillet-corrected waypoints?

$$\boxed{|W|_F = |W| + \sum_{i=2}^{N}\left[R(\pi - \varrho_i) - \frac{2R}{\tan(\varrho_i/2)}\right]} \qquad \varrho_i \triangleq \cos^{-1}\left(-\boldsymbol{q}_{i-1}^{\top}\boldsymbol{q}_i\right)$$

UAV Path Manager: Dubins Path

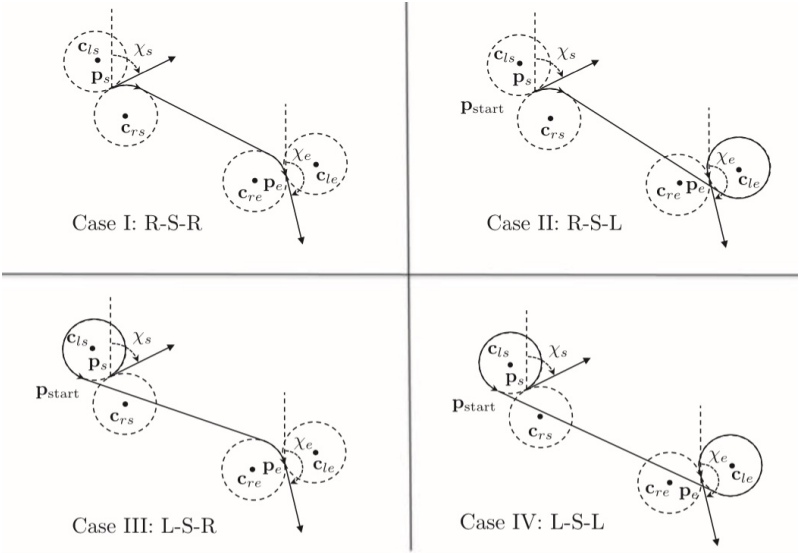
Dubins Vehicle

Vehicle kinematics with constant speed and bounded control inputs

$$\dot{p}_n = V\cos\vartheta, \quad \dot{p}_e = V\sin\vartheta, \quad \dot{\vartheta} = u, \quad u \in [-\bar{u}, \bar{u}]$$

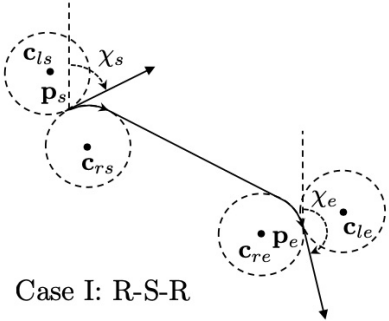
- How to obtain a time-optimal path between two configurations?
- **Dubins Paths**: A circular arc of radius R, followed by a straight line, and then another circular arc of radius R to the final configuration.
- **Radius of circular arcs**: $R = V/\bar{u} >$ Minimum turn radius
- **Vehicle's configuration**: $(\boldsymbol{p}, \chi)$ with $\boldsymbol{p}$ and χ as inertial position and course angle of UAV, respectively
- **Given configurations**: Initial $(\boldsymbol{p}_s, \chi_s)$ and final $(\boldsymbol{p}_e, \chi_e)$

UAV Path Manager: Dubins Path



UAV Path Manager: Dubins Path Length Computation

- How to compute the path length of each case?
- Need to find path lengths of all segments (circular arcs and straight lines)
- How to obtain centers of both turning circles?

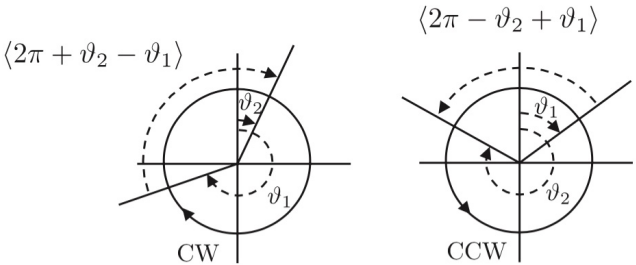


- Centers of right and left turning circles for a given position $\boldsymbol{p}$, course angle χ, and radius R

$$\boldsymbol{c}_r = \boldsymbol{p} + R\left(\cos\left(\chi + \pi/2\right), \sin\left(\chi + \pi/2\right), 0\right)^\top$$

$$\boldsymbol{c}_l = \boldsymbol{p} + R\left(\cos\left(\chi - \pi/2\right), \sin\left(\chi - \pi/2\right), 0\right)^\top$$

UAV Path Manager: Dubins Path Length Computation



- Angles $\vartheta_1, \vartheta_2 \in [0, 2\pi]$
- For CW circles,

$$|\vartheta_2 - \vartheta_1|_{CW} \triangleq \langle 2\pi + \vartheta_2 - \vartheta_1 \rangle$$

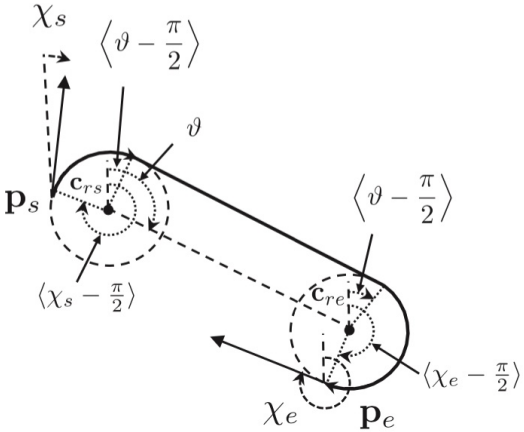
where $\langle \varphi \rangle \triangleq \varphi \bmod 2\pi$

- For CCW circles,

$$|\vartheta_2 - \vartheta_1|_{CCW} \triangleq \langle 2\pi - \vartheta_2 + \vartheta_1 \rangle$$

- Need to be careful about the direction of angle measurements CW or CCW
- How to compute the path length of an RSR kind of path?

UAV Path Manager: Dubins Path Length Computation (RSR)



- Angle between $\boldsymbol{c}_{rs}$ and $\boldsymbol{c}_{re}$: ϑ

- Distance along $\boldsymbol{c}_{rs}$

$$d_1 = R\langle 2\pi + \langle \vartheta - \pi/2 \rangle - \langle \chi_s - \pi/2 \rangle \rangle$$

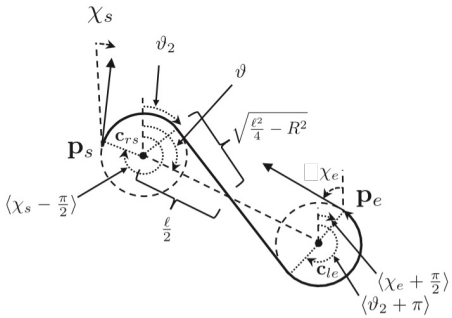
- Distance along $\boldsymbol{c}_{re}$

$$d_2 = R\langle 2\pi + \langle \chi_e - \pi/2 \rangle - \langle \vartheta - \pi/2 \rangle \rangle$$

- Total path length for RSR case

$$\begin{aligned} L_1 &= \|\boldsymbol{c}_{rs} - \boldsymbol{c}_{re}\| + d_1 + d_2 \\ &= \|\boldsymbol{c}_{rs} - \boldsymbol{c}_{re}\| \\ &\quad + R\langle 2\pi + \langle \vartheta - \pi/2 \rangle - \langle \chi_s - \pi/2 \rangle \rangle \\ &\quad + R\langle 2\pi + \langle \chi_e - \pi/2 \rangle \rangle - \langle \vartheta - \pi/2 \rangle \rangle \end{aligned}$$

UAV Path Manager: Dubins Path Length Computation (RSL)



- Angle between $\boldsymbol{c}_{rs}$ and $\boldsymbol{c}_{le}$: ϑ
- Distance between $\boldsymbol{c}_{rs}$ and $\boldsymbol{c}_{le}$: $l = \|\boldsymbol{c}_{le} - \boldsymbol{c}_{rs}\|$
- What about angle ϑ_2?

- Angle $\vartheta_2 = \vartheta - \left[\frac{\pi}{2} - \sin^{-1}\left(\frac{2R}{l}\right)\right]$
- Distance along $\boldsymbol{c}_{rs}$

$$d_3 = R\langle 2\pi + \langle \vartheta_2 \rangle - \langle \chi_s - \pi/2 \rangle\rangle$$

- Distance along $\boldsymbol{c}_{le}$

$$d_4 = R\langle 2\pi + \langle \vartheta_2 + \pi \rangle - \langle \chi_e + \pi/2 \rangle\rangle$$

- Total path length for RSL case

$$\begin{aligned} L_2 &= \sqrt{l^2 - 4R^2} + d_3 + d_4 \\ &= \sqrt{l^2 - 4R^2} + R\langle 2\pi + \langle \vartheta_2 \rangle - \langle \chi_s - \pi/2 \rangle\rangle \\ &\quad + R\langle 2\pi + \langle \vartheta_2 + \pi \rangle - \langle \chi_e + \pi/2 \rangle\rangle \end{aligned}$$

Textbooks/References

References

- Randal Beard and Timothy W. McLain, "Small Unmanned Aircraft: Theory and Practice", Princeton University Press, 2012

Thank you for your attention !!!